A Complemented-Subspace Reduction for a Rosenthal–Odell Quotient Question

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Status

partial_result (complemented subspaces of quotients of strict ℓ_1-preduals contain c_0, plus a general (WU) trichotomy and c_0-saturation criterion).

This packet is the single durable packet for the current Rosenthal–Odell target. Its promoted mathematical claim is the partial theorem below. The later sections record additional constraints on any remaining counterexample, including a general (WU)-based trichotomy. These reductions are not a full solution. A broader literature-implied answer for the isomorphic ℓ_1-predual interpretation is recorded later as context, not as the promoted claim.

Source Question

The source paper is Gasparis, *New examples of c_0-saturated Banach spaces*, arXiv:0809.1808 [1]. On page 2, Gasparis records Rosenthal's question asking whether every quotient of an ℓ_1-predual is c_0-saturated, and Odell's special case for quotients of $C(K)$ with K countable compact.

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Rosenthal ([27], [28]) has also asked another related question, namely if every quotient of an ℓ_1-predual is c_0-saturated. E. Odell [24] asked the same question for the special case of $C(K)$ spaces with K countable and compact. Note that if some of the spaces E_n given in the statement of Theorem

Partial Result

Theorem. *Let E be a real Banach space such that E^* is isometric to ℓ_1, and let X be a quotient of E. Then every complemented infinite-dimensional subspace of X contains an isomorphic copy of c_0.*

Consequently, if K is a countable compact metric space and X is a quotient of $C(K)$, every complemented infinite-dimensional subspace of X contains c_0.

Proof Intuition

The full Rosenthal–Odell problem is hard because a bad subspace of a quotient need not itself be a quotient. The complemented case removes exactly that difficulty. If Y is complemented in X, the projection $P : X \to Y$ turns Y into the range of an operator on the whole quotient.

Quotients of real L^1-preduals have Pelczynski's property (V), so if Y contains no c_0, the projection must be weakly compact. That makes Y reflexive. But in the strict ℓ_1-predual/countable $C(K)$ setting, a complemented subspace of a quotient is itself a quotient, so its dual embeds into ℓ_1; an infinite-dimensional reflexive space cannot have dual embedded in ℓ_1. Thus the complemented bad subspace cannot exist.

Proof

We use four standard facts. First, by Johnson–Zippin's theorem, in the form recalled for example in [2, Theorem 4.5], every real L^1-predual has Pelczynski's property (V). Second, property (V) passes to quotients: if $Q : E \to X$ is onto and $T : X \to Z$ is unconditionally converging, then $TQ : E \to Z$ is unconditionally converging, hence weakly compact; by the open mapping theorem this implies that T is weakly compact. Third, an operator is unconditionally converging if and only if it fixes no copy of c_0. Fourth, every subspace of ℓ_1 has the Schur property, so it contains no infinite-dimensional reflexive subspace.

Let $Q : E \to X$ be a quotient map and let $Y \subset X$ be complemented. Let $P : X \to Y$ be a bounded projection. Suppose, toward a contradiction, that Y is infinite-dimensional and contains no copy of c_0.

The projection P fixes no copy of c_0. Indeed, if there were a subspace $Z \subset X$, $Z \simeq c_0$, such that $P|_Z$ were an isomorphic embedding, then $PZ \subset Y$ would be isomorphic to c_0, contrary to the assumption on Y. Hence P is unconditionally converging. Since X is a quotient of the real L^1-predual E, the quotient-stability observation above gives property (V) for X, and therefore P is weakly compact.

But $P|_Y$ is the identity on Y. Thus $B_Y \subset P(B_X)$, up to the harmless inclusion $B_Y \subset B_X$, and B_Y is relatively weakly compact. Hence Y is reflexive.

Now $PQ : E \to Y$ is onto. Its adjoint $(PQ)^* : Y^* \to E^*$ is an isomorphic embedding. Since E^* is isometric to ℓ_1, Y^* is isomorphic to a reflexive subspace of ℓ_1. A reflexive Schur space is finite-dimensional: its closed unit ball is weakly compact and, by the Schur property, weak compactness is norm compactness. Hence Y^*, and therefore Y, is finite-dimensional, contradicting the choice of Y.

This proves that every complemented infinite-dimensional subspace of X contains c_0. If K is countable compact metric, then $C(K)^*$ is isometric to $\ell_1(K)$, so the same conclusion applies to every quotient of $C(K)$.

Literature-Implied Broad Answer

There is a broader reading of Rosenthal's phrase "ℓ_1-predual" under which one only requires X^* to be isomorphic to ℓ_1. On that broader, isomorphic reading, the quotient question has a negative answer by direct identification with later Bourgain–Delbaen/Argyros–Motakis constructions.

Argyros–Motakis [3] construct a hereditarily indecomposable $\mathscr{L}_\infty$-space $\mathfrak{X}_{\mathrm{nr}}$ which contains no c_0, no ℓ_1, and no reflexive subspace. In their quotient section they state that there are infinite-dimensional spaces X_1, X_2, X_3 such that X_2 is a quotient of X_1, X_3 is a quotient of X_2, X_1 and X_3 are reflexively saturated, X_2 contains no reflexive subspace, and all three spaces are ℓ_1-preduals with the scalar-plus-compact property. The surrounding discussion identifies the middle space with the non-c_0 $\mathscr{L}_\infty$ construction.

Thus, if "ℓ_1-predual" is interpreted isomorphically, $X_1 \twoheadrightarrow X_2$ is a quotient of an ℓ_1-predual which is not c_0-saturated, since X_2 itself contains no copy of c_0. This is a literature-implied answer and is not claimed here as a new proof. It also does not settle Odell's special problem for quotients of $C(K)$, K countable compact.

Final-Push Reduction for the Countable $C(K)$ Case

The following elementary observation is useful when trying to attack the uncomplemented part of Odell's problem.

Lemma. *Let K be countable compact metric, let X be a quotient of $C(K)$, and let $Y \subset X$ be a subspace. Then Y^* is separable. Consequently Y contains no copy of ℓ_1.*

Proof. Since X is a quotient of $C(K)$, X^* embeds isometrically into $C(K)^* \cong \ell_1(K)$. In particular X^* is separable. The restriction map $X^* \to Y^*$ is onto by Hahn–Banach, so Y^* is separable as well. If Y contained ℓ_1, then restriction from Y^* onto the dual of that copy would give a quotient map $Y^* \to \ell_\infty$. This is impossible because quotients of separable spaces are separable while ℓ_∞ is not. □

For the first hard ordinal level $C(\omega^\omega)$, Odell records a further obstruction due to Schlumprecht [4]: if X is a quotient of $C(\omega^\omega)$, then the closed linear span of any normalized weakly null sequence in X whose spreading model is equivalent to the ℓ_1 basis contains c_0. Therefore any c_0-free subspace Y of a quotient of $C(\omega^\omega)$ must be very special: it has separable dual, contains no ℓ_1, and has no normalized weakly null sequence with ℓ_1 as a spreading model.

This reduction explains why the remaining case resembles the modern Bourgain–Delbaen/Argyros–Haydon low-Szlenk pathology rather than a simple ℓ_1 obstruction. Alspach's analysis of the Bourgain–Delbaen space [8] shows that there are $\mathscr{L}_\infty$-spaces with separable dual, no c_0, and small Szlenk index; such examples stress-test any proposed proof, even though they are not known here to be quotients of countable ordinal C-spaces. Conversely, Alspach's earlier note [7] emphasizes that quotients of $C(\alpha)$, $\alpha < \omega_1$, were already subtle in the strict ℓ_1-predual setting.

Extra Push: General (WU) Trichotomy

Odell [4] says that a Banach space has property (WU) when every normalized weakly null sequence has an unconditional subsequence. His Theorem A proves that every quotient of a space with a shrinking unconditional finite-dimensional decomposition has (WU). The same paper also points out that $C(\omega^\omega)$ has (WU), but asks whether quotients preserve (WU) in this setting.

The preceding countable $C(K)$ reduction is a special case of the following general trichotomy. The point is not a new construction, but a useful packaging of Rosenthal's theorem, Odell's (WU) hypothesis, and James's unconditional-basis criteria.

Theorem (General (WU) trichotomy)**.** *Let X be an infinite-dimensional Banach space with property (WU). Then every infinite-dimensional subspace $Y \subset X$ contains a copy of ℓ_1, a copy of c_0, or an infinite-dimensional reflexive subspace.*

In particular, if X has (WU) and contains neither ℓ_1 nor c_0, then X is reflexively saturated.

Proof. Property (WU) passes to subspaces: if $Y \subset X$ and a sequence is weakly null in Y, then it is weakly null in X. Let $Y \subset X$ be infinite-dimensional. If Y contains ℓ_1 or c_0, there is nothing to prove. Assume that Y contains neither ℓ_1 nor c_0.

If Y is reflexive, again there is nothing to prove. Suppose that Y is not reflexive. By Eberlein–Smulian, choose a bounded sequence in Y with no weakly convergent subsequence. Since Y contains no ℓ_1, Rosenthal's ℓ_1-theorem [5] gives a weakly Cauchy subsequence (y_n). This subsequence is not weakly convergent, by the choice of the original sequence, and hence it is not norm Cauchy.

Thus there are indices $n_1 < m_1 < n_2 < m_2 < \cdots$ and a number $\delta > 0$ such that

$$\|y_{m_j} - y_{n_j}\| \geq \delta \qquad (j \in \mathbb{N}).$$

Set

$$z_j = \frac{y_{m_j} - y_{n_j}}{\|y_{m_j} - y_{n_j}\|}.$$

For every $f \in Y^*$, the scalar sequence $f(y_n)$ is Cauchy, so $f(z_j) \to 0$. Hence (z_j) is normalized and weakly null in Y. By (WU), some subsequence (z_{j_k}) is unconditional basic.

Let $Z = [z_{j_k}]$. Since $Z \subset Y$, the space Z contains neither ℓ_1 nor c_0. James's characterizations for unconditional bases now apply [6]: an unconditional basis is shrinking if its closed span contains no ℓ_1, and boundedly complete if its closed span contains no c_0. Therefore (z_{j_k}) is both shrinking and boundedly complete, so Z is reflexive. This gives the required infinite-dimensional reflexive subspace of Y. □

Corollary (No-ℓ_1 (WU) criterion for c_0-saturation)**.** *Let X have property (WU) and contain no copy of ℓ_1. Then every infinite-dimensional subspace of X contains c_0 or an infinite-dimensional reflexive subspace. Consequently,*

$$X \textit{ is } c_0\textit{-saturated} \iff X \textit{ contains no infinite-dimensional reflexive subspace.}$$

Proof. The first assertion is the trichotomy with the ℓ_1 alternative removed. If X contains an infinite-dimensional reflexive subspace, then X is not c_0-saturated, since subspaces of reflexive spaces are reflexive and c_0 is not reflexive. Conversely, if X is not c_0-saturated, choose an infinite-dimensional subspace $Y \subset X$ containing no c_0. The first assertion gives an infinite-dimensional reflexive subspace of Y, and hence of X. □

Corollary (Quotients of shrinking unconditional FDD spaces)**.** *Let X be a quotient of a Banach space with a shrinking unconditional finite-dimensional decomposition. Then every infinite-dimensional subspace of X contains c_0 or an infinite-dimensional reflexive subspace. In particular, X is c_0-saturated if and only if X contains no infinite-dimensional reflexive subspace.*

Proof. By Odell's theorem [4], X has property (WU). The dual of the source space is separable, so X^*, which embeds into that dual, is separable. Therefore X contains no copy of ℓ_1. Apply the preceding corollary. □

For the countable compact $C(K)$ target, the same criterion applies whenever the quotient X itself is known to have (WU): the earlier lemma gives X^* separable, hence no ℓ_1, and the no-ℓ_1 (WU) criterion says that X is c_0-saturated exactly when X contains no infinite-dimensional reflexive subspace. More locally, any c_0-free subspace Y of a quotient of countable $C(K)$ which has (WU) must contain an infinite-dimensional reflexive subspace.

For quotients of $C(\omega^\omega)$, this combines with Schlumprecht's observation as follows. A hypothetical c_0-free subspace with no infinite-dimensional reflexive subspace must fail (WU): it contains a normalized weakly null sequence with no unconditional subsequence. At the same time, none of its normalized weakly null sequences can have an ℓ_1 spreading model, because such a sequence would generate a closed span containing c_0. Thus any no-reflexive counterexample must be a very specific Maurey–Rosenthal-type obstruction inside a quotient of $C(\omega^\omega)$, not merely a Bourgain–Delbaen-style no-c_0 space with separable dual.

Limitations

This is not a full solution of the Rosenthal–Odell question. The open problem asks about every infinite-dimensional subspace of the quotient. The argument above uses the projection onto Y, so it only rules out complemented c_0-free subspaces. Any counterexample to the strict ℓ_1-predual or countable compact $C(K)$ version must therefore contain its c_0-free subspace in an uncomplemented way.

A naive Cantor–Bendixson induction would quotient by the first derivative kernel $C_0(K \setminus K') \cong c_0$. The induction would close if the image of this kernel in every quotient were closed or at least c_0-saturated. This is exactly the dangerous point: bounded operators from c_0 may have nonclosed range with arbitrary separable closure, so the closure of the kernel image can in principle hide a c_0-free low-Szlenk subspace. Any full proof must rule out that behavior using the specific ordinal $C(K)$ structure, not just the abstract exact sequence.

Another tempting shortcut also fails: although Y^* is a quotient of a subspace of ℓ_1, it need not have the Schur property. Quotients of ℓ_1 can be arbitrary separable Banach spaces. Thus the final-push reduction cannot rule out reflexive or James-type subspaces merely from the dual embedding into ℓ_1; it only rules out ℓ_1 itself and, in the $C(\omega^\omega)$ case, weakly null ℓ_1-spreading-model routes.

The extra-push (WU) trichotomy above also does not solve the problem. It shows that, under (WU) and no ℓ_1, failure of c_0-saturation is equivalent to the presence of an infinite-dimensional reflexive subspace. Odell explicitly lists quotient preservation of (WU) as open in the relevant $C(\omega^\omega)$ neighborhood, and the existence of reflexive subspaces in quotients of countable $C(K)$ spaces is not ruled out by the present argument.

Verification Notes

The proof is qualitative and uses no computation. The key review points are the real-scalar convention in Johnson–Zippin's property (V) theorem and the standard equivalence between unconditionally converging operators and operators fixing no copy of c_0. The countable compact $C(K)$ specialization is strict: $C(K)^* \cong \ell_1(K)$ isometrically.

For the literature-implied broad answer, review should check the convention for "ℓ_1-predual" and the identification of the middle quotient X_2 in [3] with the no-c_0 Argyros–Motakis space. That contextual answer should not be used to mark the countable compact $C(K)$ problem solved.

For the final-push reduction, review should check two points. First, the separable-dual obstruction uses only $X^* \subset \ell_1(K)$ and Hahn–Banach restriction, and deliberately does not claim that Y^* has the Schur property. Second, Schlumprecht's observation as recorded in [4] is used here only for quotients of $C(\omega^\omega)$ and only to rule out ℓ_1-spreading-model weakly null sequences inside a hypothetical c_0-free subspace.

For the extra-push (WU) trichotomy, review should check the inheritance of (WU) by subspaces, the use of Rosenthal's ℓ_1-theorem to obtain a nontrivial weakly Cauchy sequence, and James's shrinking/boundedly-complete characterizations for unconditional bases. The criterion assumes (WU) for the bad subspace or quotient; it does not assert that arbitrary quotients of $C(\omega^\omega)$ have (WU). A bounded local-index and web search found Odell's source theorem and related problem context, but not an exact pre-existing statement of the packaged trichotomy; since the proof is a direct combination of standard theorems, novelty should be treated as a useful derived reduction rather than as a claimed independent classical theorem.

References

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